Order Strepsiptera

Strepsipterans

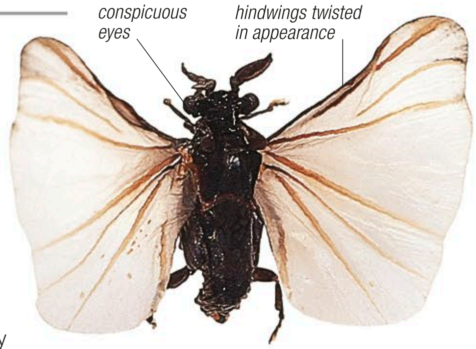
With their small size, parasitic lifestyle, and extreme sex differences, strepsipterans make up one of the most unusual orders of insects. Males usually have wings, the hindwings often being twisted, but females are grublike, and spend their lives lodged in the bodies of larger insects, such as bugs, bees, or wasps. There are about 580 species, in 8 families, and all undergo complete metamorphosis.

Family Stylopidae

Stylopids

Occurrence 268 spp. worldwide; in vegetated habitats where hosts are found, females in bees and wasps

Male stylopids are small and dark, with bulging eyes and bodies that are 1/64–5/32 in (0.5–4 mm) long. Like all strepsipterans, their forewings are tiny and their hindwings relatively large and fanlike. Females have neither legs nor wings, and never leave the body of the host insect. After mating, thousands of eggs hatch inside the female; some species may lay up to 70,000 eggs. The active, tiny, 6-legged larvae (triungulin) leave the female through a special passage and crawl onto flowers to wait for the next host. They may cling to the host's body or be ingested with nectar. In many cases, once inside the host's nest, they leave the body or are regurgitated and proceed to parasitize the eggs or larvae. The sexual organs of the parasitized host also degenerate.



STYLOPS SP.
Males of these species have a highly distinctive shape, with branched antennae and the order's characteristic berrylike eyes.

Order Mecoptera

Scorpionflies

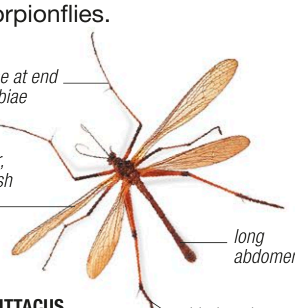
The males of some species in this order have a scorpionlike, slender abdomen—hence the common name. Metamorphosis is complete. There are 9 families and 550 species of scorpionflies.

Family Bittacidae

Hangingflies

Occurrence 170 spp. in the Southern Hemisphere; in damp, wooded or well-vegetated, shady areas

Hangingflies use their front legs to hang from vegetation and hindlegs to catch prey. The adults are 1/2–1 1/2 in (1.5–4 cm) long. Eggs are laid in soil.



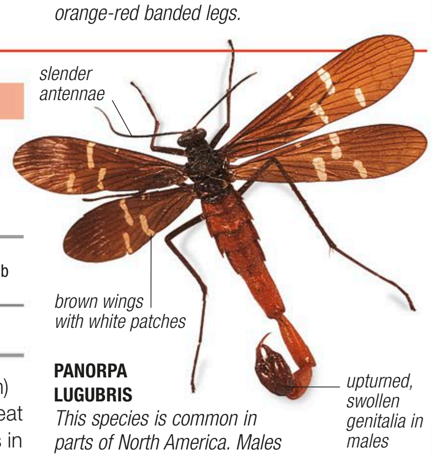
HARPOBITTACUS AUSTRALIS
This is an Australian species with brownish wings and orange-red banded legs.

Family Panorpidae

Common scorpionflies

Occurrence 360 spp. mostly in the Northern Hemisphere; among vegetation in woodland and scrub

These insects are 11/32–1 in (0.9–2.5 cm) long, often with mottled wings. Adults eat nectar, fruit, and insects. They lay eggs in soil, where hatched larvae pupate.



PANORPA LUGUBRIS
This species is common in parts of North America. Males are larger than the females.

Order Siphonaptera

Fleas

Fleas are brown, shiny, wingless insects, superbly adapted for a parasitic lifestyle feeding on mammalian blood. Their bodies are flattened sideways, which helps them to slip through fur, and they are very tough, which makes them hard to kill. They leap onto suitable host animals using their enlarged hindlegs, which store energy in special pads of rubberlike protein. There are over 2,500 species (18 families) of fleas; all undergo complete metamorphosis. Larvae scavenge on debris, dried blood, and adult flea excrement.

Family Pulicidae

Common fleas

Occurrence 200 spp. worldwide; on mammalian hosts, in lairs, burrows

Common fleas parasitize humans and a wide range of other mammals, including dogs, cats, and other carnivores, as well as hedgehogs, rabbits, hares, and rodents. Some of them have a broad host range, occurring on up to 30 different animal species. Common fleas are 1/32–5/16 in (1–8 mm) long and often have combs of backward-pointing bristles on their cheeks—features that are an important aid in species identification. Adult females lay eggs as they feed, dropping them into their hosts' nests, burrows, or bedding, where they hatch to produce pale, wormlike larvae that have tiny, biting jaws. The larvae typically live for 2–3 weeks, before pupating in a minute silk cocoon. Once they have become adult, the fleas remain inside their cocoons for weeks or months until they sense vibrations from a nearby animal. They then emerge from their cocoons and jump aboard their prospective host, attracted by its body heat. However, if they fail to find a host, or become dislodged, they can survive for a long time without food. Many species of common fleas, such as the cat flea, dog flea, and human flea, are of medical importance. Their bite causes severe itching and allergic reactions, as well as spreads diseases and parasitic worms. The bacterium that caused bubonic plague in medieval Europe was carried by various types of rat flea.

SPILOPSYLLUS CUNICULI
Commonly known as the rabbit flea, this species is widespread in the Northern Hemisphere. It carries the rabbit disease myxomatosis.

PULEX IRRITANS
This species was once the most common flea attacking humans; today, cat and dog fleas are more dominant. It also attacks pigs, goats, and badgers, and can be a carrier of bubonic plague.

CEDISPSYLLA SIMPLEX
This species is found on cottontail rabbits in North America.

CTENOCEPHALIDES CANIS
Very common in human habitations the world over, the dog flea is reddish brown, with a head that is more domed than that of the cat flea. It is a carrier of the dog tapeworm, which can also affect cats and humans.

CTENOCEPHALIDES FELIS
The cat flea is a common domestic species found in temperate regions. A cat may have only a few adult fleas feeding on it but its bed supports thousands of larvae. Hungry cat fleas can jump up to 13 1/2 in (34 cm) and readily bite humans.